

SAE4D01_TP1_GNMIC_MALOT_VALENTIN

Synthèse

Durant ce TP, j'ai utilisé `gnmic`, un client en ligne de commande qui implémente le protocole **gNMI** (gRPC Network Management Interface). `gNMI` repose sur **gRPC** comme protocole de transport et sur les modèles de données YANG (OpenConfig dans le cadre de la SAE) pour structurer les informations échangées. L'avantage de cette approche est la standardisation : les mêmes chemins YANG fonctionnent sur des équipements de constructeurs différents (Arista, Juniper, Nokia), ce qui facilite l'automatisation dans des environnements hétérogènes.

Aussi, durant le TP, j'ai utilisé 3 instructions principales avec `gNMIC`, `get` pour lire une valeur à un instant donné, `set` pour modifier la configuration, et `subscribe` pour recevoir des données périodiquement (un peu comme un abonnement MQTT).

Il existe aussi deux modes : `sample` qui permet un échantillonnage périodique mais aussi `on-change` qui déclenche une notification lors d'un changement de valeur. Contrairement au protocole SNMP, `gNMI` permet à la fois la lecture, l'écriture et la lecture en direct de données dans un seul protocole.

Aussi,, `gnmic` intègre nativement un output vers Prometheus. Il est possible comme vu en fin de TP que les souscriptions `gNMI` qui sont définies dans `gnmic.yaml` soient collectées en continu et exposées sur un endpoint `/metrics` (port 9273). Dans le TP, j'ai par exemple configuré Prometheus pour qu'il scrape cet endpoint toutes les 15 secondes.

En théorie, il est donc possible d'intégrer `gNMIC` à une stack d'observabilité classique Prometheus + Grafana.

TP1 : Télémétrie des équipements réseau

1. Installation des dépendances et configurations initiales.

```
mkdir -p /root/TP_GNMIC  
cd /root/TP_GNMIC
```

```
curl -fsSL https://pyenv.run | bash
```

J'ai modifié le `.bashrc` en y ajoutant le contenu suivant :

```
export PYENV_ROOT="$HOME/.pyenv"  
[[ -d $PYENV_ROOT/bin ]] && export PATH="$PYENV_ROOT/bin:$PATH"  
eval "$(pyenv init - bash)"  
eval "$(pyenv virtualenv-init -)"
```

Installation des dépendances :

```
sudo apt install -y \  
    make \  
    build-essential \  
    libssl-dev \  
    zlib1g-dev \  
    libbz2-dev \  
    libreadline-dev \  
    libsqlite3-dev \  
    wget \  
    curl \  
    llvm \  
    libncurses5-dev \  
    libncursesw5-dev \  
    xz-utils \  
    tk-dev \  
    libffi-dev \  
    libbz2-dev
```

```
liblzma-dev \  
python3-openssl \  
libssh-dev \  
openvswitch-switch \  
sshpas
```

Puis

```
pyenv install -l | grep 3.11  
pyenv install 3.11.11  
pyenv virtualenv 3.11.11 netlab  
pyenv activate netlab
```

Puis

```
bash -c "$(curl -sL https://get-gnmic.openconfig.net)"
```

Puis :

```
cat << EOF > ./topology.yml  
message: >  
  2 routeurs Arista avec BGP et OSPF et 2 clients Linux.  
  
tools:  
  # graphe de la maquette voir http://127.0.0.1/graphite  
  graphite:  
  
module: [ ospf, bgp ]  
defaults.device: eos  
provider: clab  
  
nodes:  
  ceos1:  
    image: registry.iutbeziers.fr/iutimages/ceos:4.33.1F  
    ospf.area: 0  
    bgp.as: 65100  
    mgmt.ipv4: 192.168.121.101
```

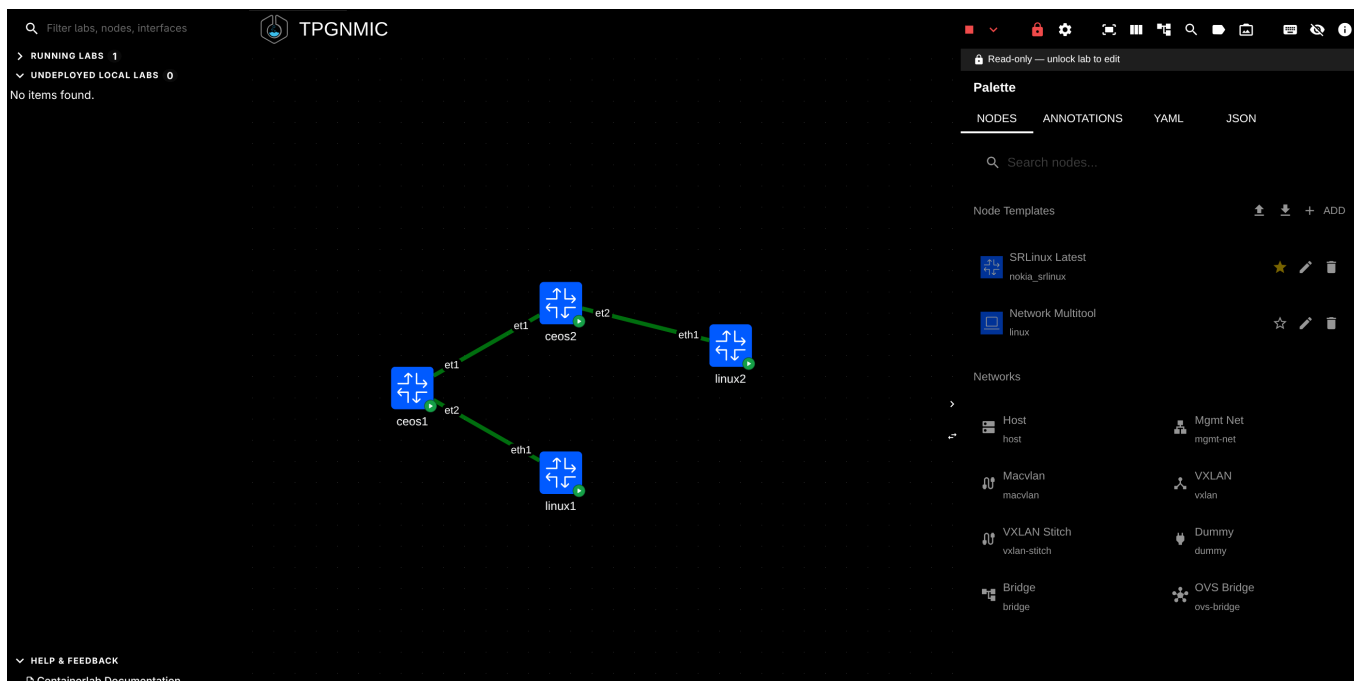
```
ceos2:
  image: registry.iutbeziers.fr/iutimages/ceos:4.33.1F
  ospf.area: 0
  bgp.as: 65200
  mgmt.ipv4: 192.168.121.102
linux1:
  device: linux
  image: registry.iutbeziers.fr/iutimages/network-
multitool:mai2026
  mgmt.ipv4: 192.168.121.103
linux2:
  device: linux
  image: registry.iutbeziers.fr/iutimages/network-
multitool:mai2026
  mgmt.ipv4: 192.168.121.104

links:
- ceos1-ceos2
- ceos1-linux1
- ceos2-linux2
EOF

netlab create topology.yml
netlab up topology.yml
netlab report addressing
netlab connect ceos1

gnmic -a 192.168.121.102:6030 -u admin -p admin --insecure
capabilities
```

Afin de visualiser la topologie du réseau, j'ai utilisé l'extension containerlab de Visual Studio Code et l'extension Remote SSH qui m'a permis de me connecter à la machine virtuelle pour simuler un environnement local.



4. Utilisation de GNMIC

4.1. Configuration de GNMIC

```
(netlab) root@debian13vm:~/TP_GNMIC# gnmic -a
192.168.121.102:6030 -u admin -p admin --insecure capabilities
gNMI version: 0.7.0
supported models:
  - openconfig-network-instance, OpenConfig working group, 4.4.1
  - openconfig-platform-transceiver, OpenConfig working group,
0.14.0
  - openconfig-bfd, OpenConfig working group, 0.3.0
  - openconfig-if-aggregate, OpenConfig working group, 2.4.4
  - openconfig-system-grpc, OpenConfig working group, 1.1.0
  - openconfig-interfaces, OpenConfig working group, 3.7.1
  - openconfig-ospf-types, OpenConfig working group, 0.1.3
  - openconfig-ospfv2, OpenConfig working group, 0.5.2
  - ietf-netconf, IETF NETCONF (Network Configuration) Working
Group,
  - arista-nat-nat-dynamic, Arista Networks
```

```
<http://arista.com/>, 0.1.0
- arista-bgp-augments, Arista Networks, Inc., 2.27.0
- arista-terminal-device-notsupported-deviations, Arista
Networks, Inc.,

(...)
```

1. Créez un fichier de configuration gnmic.yaml dans le répertoire projet afin d'éviter de resaisir les paramètres de connexion.

J'ai cherché la syntaxe du fichier de configuration sur le site internet de gNMIC :
https://gnmic.openconfig.net/user_guide/configuration_file/

L'exemple fourni est le suivant :

```
# gNMI target address; CLI flag `--address`
address: "10.0.0.1:57400"
# gNMI target user name; CLI flag `--username`
username: admin
# gNMI target user password; CLI flag `--password`
password: NokiaSrl1!
# connection mode; CLI flag `--insecure`
insecure: true
# log file location; CLI flag `--log-file`
log-file: /tmp/gnmic.log
```

Je l'ai adapté à mon environnement :

```
(netlab) root@debian13vm:~/TP_GNMIC# cat gnmic.yaml
address: "192.168.121.102:6030"
username: admin
password: admin
insecure: true
log-file: /tmp/gnmic.log
```

Ainsi, le résultat est identique au précédent :

```
(netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml
capabilities
gNMI version: 0.7.0
supported models:
  - arista-exp-eos-vxlan-config, Arista Networks
    <http://arista.com/>, 0.2.1
  - openconfig-icmpv6-types, OpenConfig working group, 0.1.1
  - openconfig-isis, OpenConfig working group, 1.7.0
  - arista-exp-eos, Arista Networks <http://arista.com/>,
  - arista-gnoi-cert, Arista Networks, Inc.,
    https://github.com/openconfig/gnoi
  - openconfig-platform-controller-card, OpenConfig working
    group, 0.2.0
  - arista-macsec-deviations, Arista Networks, Inc., 1.0.0
  - arista-ale-counters-open-config-specialized-counters, Arista
    Networks <http://arista.com/>, 1.0.0
  - openconfig-local-routing, OpenConfig working group, 2.0.1
  - arista-routing-policy-notsupported-deviations, Arista
    Networks, Inc.,

(...)
```

2. Récupérer les capacités du routeur ceos1

J'ai modifié l'adresse du routeur dans le fichier de configuration `gnmic.yaml` par celle fournie pour EOS1 dans le fichier `topology.yaml` :

```
address: "192.168.121.101:6030"
```

J'ai lancé la commande suivante :

```
(netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml
capabilities
gNMI version: 0.7.0
supported models:
  - openconfig-telemetry-types, OpenConfig working group, 0.4.2
  - openconfig-mpls-sr, OpenConfig working group, 3.0.1
```

```
- openconfig-terminal-device, OpenConfig working group, 1.9.2
- arista-sand-hardware-drop-counters, Arista Networks
<http://arista.com/>, 1.0.0
- arista-telemetry-augments, Arista Networks, Inc., 1.0.0
- openconfig-platform-cpu, OpenConfig working group, 0.1.1
- arista-ale-counters-open-config-specialized-counters, Arista
Networks <http://arista.com/>, 1.0.0
- arista-ale-counters-open-config-hardware-drop-counters,
Arista Networks <http://arista.com/>, 1.0.0
- arista-srte-augments, Arista Networks, Inc., 1.1.1

(...)
```

3. Récupérer l'état de l'interface Ethernet1

```
gnmic --config gnmic.yaml get --path
"/interfaces/interface[name=Ethernet1]/state"
```



```
(netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml get --path "/interfaces/interface[name=Ethernet1]/state"
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781250156810637852,
    "time": "2026-06-12T09:42:36.810637852+02:00",
    "updates": [
      {
        "Path": "interfaces/interface[name=Ethernet1]/state",
        "values": {
          "interfaces/interface/state": {
            "arista-intf-augments:inactive": false,
            "openconfig-interfaces:admin-status": "UP",
            "openconfig-interfaces:counters": {
              "carrier-transitions": "4",
              "in-broadcast-pkts": "4",
              "in-discards": "0",
              "in-errors": "1",
              "in-fcs-errors": "0",
              "in-multicast-pkts": "67",
              "in-octets": "23105",
              "in-pkts": "149",
              "in-unicast-pkts": "78",
              "out-broadcast-pkts": "0",
              "out-discards": "0",
              "out-errors": "0",
              "out-multicast-pkts": "0",
              "out-octets": "0",
              "out-pkts": "0",
              "out-unicast-pkts": "0"
            },
            "openconfig-interfaces:description": "ceos1 -> ceos2 [external]",
            "openconfig-interfaces:ifindex": 1,
            "openconfig-interfaces:last-change": "1781248444629157066",
            "openconfig-interfaces:management": false,
            "openconfig-interfaces:mtu": 0,
            "openconfig-interfaces:name": "Ethernet1",
            "openconfig-interfaces:oper-status": "UP",
            "openconfig-interfaces:type": "iana-if-type:ethernetCsmacd",
            "openconfig-platform-port:hardware-port": "Ethernet1-Port",
            "openconfig-platform-transceiver:transceiver": "Ethernet1"
          }
        }
      }
    ]
  }
]
```

4. Récupérer les compteurs de paquets de l'interface Ethernet1

```
gnmic --config gnmic.yaml get --path
"/interfaces/interface[name=Ethernet1]/state/counters"
```

```
(netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml get --path "/interfaces/interface[name=Ethernet1]/state/counters"
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781250202379842443,
    "time": "2026-06-12T09:43:22.379842443+02:00",
    "updates": [
      {
        "Path": "interfaces/interface[name=Ethernet1]/state/counters",
        "values": {
          "interfaces/interface/state/counters": {
            "openconfig-interfaces:carrier-transitions": "4",
            "openconfig-interfaces:in-broadcast-pkts": "4",
            "openconfig-interfaces:in-discards": "0",
            "openconfig-interfaces:in-errors": "1",
            "openconfig-interfaces:in-fcs-errors": "0",
            "openconfig-interfaces:in-multicast-pkts": "69",
            "openconfig-interfaces:in-octets": "23864",
            "openconfig-interfaces:in-pkts": "154",
            "openconfig-interfaces:in-unicast-pkts": "81",
            "openconfig-interfaces:out-broadcast-pkts": "0",
            "openconfig-interfaces:out-discards": "0",
            "openconfig-interfaces:out-errors": "0",
            "openconfig-interfaces:out-multicast-pkts": "0",
            "openconfig-interfaces:out-octets": "0",
            "openconfig-interfaces:out-pkts": "0",
            "openconfig-interfaces:out-unicast-pkts": "0"
          }
        }
      }
    ]
  }
]
```

5. Récupérez le "neighbor" BGP

```
gnmic --config gnmic.yaml get --path "/network-
instances/network-
instance[name=default]/protocols/protocol[identifiant=BGP]
[name=BGP]/bgp/neighbors"
```

6. Récupérez l'état de la session BGP :

```
(netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml
get --path "/network-instances/network-
instance[name=default]/protocols/protocol[identifiant=BGP]
[name=BGP]/bgp/neighbors/neighbor/state/session-state"
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781250364094627657,
    "time": "2026-06-12T09:46:04.094627657+02:00",
    "updates": [
      {
        "Path": "network-instances/network-
instance[name=default]/protocols/protocol[identifiant=BGP]
```

```
[name=BGP]/bgp/neighbors/neighbor[neighbor-  
address=10.1.0.2]/state/session-state",  
    "values": {  
        "network-instances/network-  
instance/protocols/protocol/bgp/neighbors/neighbor/state/session  
-state": "ESTABLISHED"  
    }  
}  
]  
}  
]
```

7. Récupérez toutes les informations de l'équipement.

```
gnmic --config gnmic.yaml get --path "/" | more
```

La commande produit une output trop volumineuse à coller dans le compte rendu, voici un extrait (capture d'écran) :

```
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781250447476759565,
    "time": "2026-06-12T09:47:27.476759565+02:00",
    "updates": [
      {
        "Path": "",
        "values": {
          "": {
            "arista-exp-eos:arista": {
              "eos": {
                "arista-exp-eos-igmpsnooping:bridging": {
                  "igmpsnooping": {
                    "config": {}
                  }
                },
                "arista-exp-eos-mlag:mlag": {
                  "config": {
                    "dual-primary-action": "action-none",
                    "reload-delay-mlag-configured": false
                  }
                },
                "arista-exp-eos-multicast:routing": {
                  "multicast": {
                    "routeconfig": {
                      "static": {
                        "vrfConfig": [
                          {
                            "vrfName": "default"
                          }
                        ]
                      }
                    }
                  }
                },
                "arista-exp-eos-qos:qos": {
                  "arista-exp-eos-qos-acl-config:acl": {
                    "input": {
                      "cli": {
                        "cmapType": [

```

8. Récupérez l'état de la mémoire de l'équipement.

```
gnmic --config gnmic.yaml get --path "/system/memory/state"
```

Output :

```
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781250519090873909,
    "time": "2026-06-12T09:48:39.090873909+02:00",
    "updates": [
      {
        "Path": "system/memory/state",
        "values": {
          "system/memory/state": {
            "openconfig-system:free": "3895001088",

```

```

        "openconfig-system:physical": "8332857344",
        "openconfig-system:reserved": "8181923840",
        "openconfig-system:used": "4437856256"
    }
}
}
]
}
]

```

9. Récupérez le nombre de CPU de l'équipement au format "flat"

```

gnmic --config gnmic.yaml get \
  --path "/components/component/state" \
  --format flat | grep -i cpu

```

```

/components/component[name=CPU0]/state/openconfig-platform:description: Intel(R) Core(TM) Ultra 7 258V
/components/component[name=CPU0]/state/openconfig-platform:mfg-name: GenuineIntel
/components/component[name=CPU0]/state/openconfig-platform:name: CPU0
/components/component[name=CPU0]/state/openconfig-platform:parent: Chassis
/components/component[name=CPU0]/state/openconfig-platform:type: openconfig-platform-types:CPU
/components/component[name=CPU1]/state/openconfig-platform:description: Intel(R) Core(TM) Ultra 7 258V
/components/component[name=CPU1]/state/openconfig-platform:mfg-name: GenuineIntel
/components/component[name=CPU1]/state/openconfig-platform:name: CPU1
/components/component[name=CPU1]/state/openconfig-platform:parent: Chassis
/components/component[name=CPU1]/state/openconfig-platform:type: openconfig-platform-types:CPU
/components/component[name=CPU2]/state/openconfig-platform:description: Intel(R) Core(TM) Ultra 7 258V
/components/component[name=CPU2]/state/openconfig-platform:mfg-name: GenuineIntel
/components/component[name=CPU2]/state/openconfig-platform:name: CPU2
/components/component[name=CPU2]/state/openconfig-platform:parent: Chassis
/components/component[name=CPU2]/state/openconfig-platform:type: openconfig-platform-types:CPU
/components/component[name=CPU3]/state/openconfig-platform:description: Intel(R) Core(TM) Ultra 7 258V
/components/component[name=CPU3]/state/openconfig-platform:mfg-name: GenuineIntel
/components/component[name=CPU3]/state/openconfig-platform:name: CPU3
/components/component[name=CPU3]/state/openconfig-platform:parent: Chassis
/components/component[name=CPU3]/state/openconfig-platform:type: openconfig-platform-types:CPU
/components/component[name=CPU4]/state/openconfig-platform:description: Intel(R) Core(TM) Ultra 7 258V
/components/component[name=CPU4]/state/openconfig-platform:mfg-name: GenuineIntel
/components/component[name=CPU4]/state/openconfig-platform:name: CPU4
/components/component[name=CPU4]/state/openconfig-platform:parent: Chassis
/components/component[name=CPU4]/state/openconfig-platform:type: openconfig-platform-types:CPU
/components/component[name=CPU5]/state/openconfig-platform:description: Intel(R) Core(TM) Ultra 7 258V
/components/component[name=CPU5]/state/openconfig-platform:mfg-name: GenuineIntel
/components/component[name=CPU5]/state/openconfig-platform:name: CPU5
/components/component[name=CPU5]/state/openconfig-platform:parent: Chassis
/components/component[name=CPU5]/state/openconfig-platform:type: openconfig-platform-types:CPU

```

Je constate sur la capture d'écran ci-dessus que mon routeur possède 6 CPU numérotés de CPU0 à CPU5.

10. Explorer l'API avec gnmic prompt

```

gnmic> get --config gnmic.yaml --path "/system/memory/state"
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781250678048202536,
    "time": "2026-06-12T09:51:18.048202536+02:00",
    "updates": [
      {
        "Path": "system/memory/state",
        "values": {
          "system/memory/state": {
            "openconfig-system:free": "3908059136",
            "openconfig-system:physical": "8332857344",
            "openconfig-system:reserved": "8169103360",
            "openconfig-system:used": "4424798208"
          }
        }
      }
    ]
  }
]
gnmic>

```

capabilities	query targets gnmic capabilities
collect	collect gNMI telemetry from targets
completion	generate completion script
diff	run a diff comparison between targets
generate	generate paths or JSON/YAML objects from YANG
get	run gnmic get on targets
getset	run gnmic get then set on targets
help	Help about any command
listen	listens for telemetry dialout updates from the node
output	manipulate configured outputs

Autre exemples :

```

get --path /interfaces
get --path /system

```

11. Souscrivez à l'état des compteurs de l'interface Ethernet1 en échantillonnant toutes les 5 secondes.

```

gnmic --config gnmic.yaml subscribe \
  --path "/interfaces/interface[name=Ethernet1]/state/counters" \
  --mode stream \
  --stream-mode sample \
  --sample-interval 5s

```

```

{
  "source": " 192.168.121.101:6030",
  "subscription-name": "default-1781250896",
  "timestamp": 1781248450773472234,
  "time": "2026-06-12T09:14:10.773472234+02:00",
  "updates": [
    {
      "Path":
"interfaces/interface[name=Ethernet1]/state/counters/in-

```

```
broadcast-pkts",
    "values": {
        "interfaces/interface/state/counters/in-broadcast-pkts":
4        }
    }
]
}
{
    "source": " 192.168.121.101:6030",
    "subscription-name": "default-1781250896",
    "timestamp": 1781248417213905160,
    "time": "2026-06-12T09:13:37.21390516+02:00",
    "prefix":
"interfaces/interface[name=Ethernet1]/state/counters",
    "updates": [
        {
            "Path": "in-discards",
            "values": {
                "in-discards": 0
            }
        },
        {
            "Path": "in-fcs-errors",
            "values": {
                "in-fcs-errors": 0
            }
        },
        {
            "Path": "out-broadcast-pkts",
            "values": {
                "out-broadcast-pkts": 0
            }
        },
        {
            "Path": "out-discards",
            "values": {
                "out-discards": 0
            }
        }
    ]
}
```

```
    },
    {
      "Path": "out-errors",
      "values": {
        "out-errors": 0
      }
    },
    {
      "Path": "out-multicast-pkts",
      "values": {
        "out-multicast-pkts": 0
      }
    },
    {
      "Path": "out-octets",
      "values": {
        "out-octets": 0
      }
    },
    {
      "Path": "out-pkts",
      "values": {
        "out-pkts": 0
      }
    },
    {
      "Path": "out-unicast-pkts",
      "values": {
        "out-unicast-pkts": 0
      }
    }
  ]
}
{
  "source": " 192.168.121.101:6030",
  "subscription-name": "default-1781250896",
  "timestamp": 1781248427390993932,
  "time": "2026-06-12T09:13:47.390993932+02:00",
  "updates": [
```



```

    {
      "Path":
"interfaces/interface[name=Ethernet1]/state/counters/in-errors",
      "values": {
        "interfaces/interface/state/counters/in-errors": 1
      }
    }
  ]
}

```

12. Récupérer les routes BGP

```

root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml get \
  --path "/network-instances/network-
instance[name=default]/protocols/protocol[identifiant=BGP]
[name=BGP]/bgp/rib"
[
  {
    "source": " 192.168.121.101:6030",
    "timestamp": 1781251013618974218,
    "time": "2026-06-12T09:56:53.618974218+02:00"
  }
]

```

13. Récupérer les processus de l'équipement. Quel est leur nombre?

```

gnmic --config gnmic.yaml get \
  --path "/system/processes"

```

Voici le résultat que j'ai obtenu en lançant la commande en ajoutant `| more` :

```

○ (netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml get \
--path "/system/processes" | more
[
{
  "source": " 192.168.121.101:6030",
  "timestamp": 1781268833303111137,
  "time": "2026-06-12T14:53:53.303111137+02:00",
  "updates": [
    {
      "Path": "system/processes",
      "values": {
        "system/processes": {
          "openconfig-system:process": [
            {
              "pid": "1",
              "state": {
                "args": [
                  "systemd.setenv=EOS_PLATFORM=ceoslab systemd.setenv=SKIP_ZEROTOUCH_BARRIER_IN_SYSDBINIT=1 systemd.setenv=CLAB_LABEL_CLAB_NODE_LONGNAME=clab-TPGNMIC-ceos
1 systemd.setenv=CLAB_LABEL_CLAB_NODE_KIND=ceos systemd.setenv=CLAB_LABEL_CLAB_NODE_GROUP=as65100 systemd.setenv=container=docker systemd.setenv=ETBA=1 systemd.setenv=NO_PR
OXY=localhost,127.0.0.1,::1,* local,192.168.121.0/24,192.168.121.101,192.168.121.102,192.168.121.103,192.168.121.104,ceos1,ceos2,linux1,linux2 systemd.setenv=CLAB_LABEL_CLA
B_NODE_NAME=ceos1 systemd.setenv=CEOS=1 systemd.setenv=CLAB_LABEL_CLAB_TOPO_FILE=/root/TP_GNMIC/clab.yaml systemd.setenv=CLAB_MGMT_VRF=management systemd.setenv=no_proxy=loc
alhost,127.0.0.1,::1,* local,192.168.121.0/24,192.168.121.101,192.168.121.102,192.168.121.103,192.168.121.104,ceos1,ceos2,linux1,linux2 systemd.setenv=CLAB_LABEL_CLAB_NODE

```

Chaque processus est affiché à la suite au format JSON.

14. Récupérer la description de l'interface Ethernet1, changez-la en “effaçable” puis supprimez-la

Voici les commandes qui permettent de :

1. Récupérer la description de l'interface Ethernet1 :

```

gnmic --config gnmic.yaml get \
  --path
"/interfaces/interface[name=Ethernet1]/config/description"

```

2. Changer la description par effaçable :

```

gnmic --config gnmic.yaml set \
  --update-path
"/interfaces/interface[name=Ethernet1]/config/description" \
  --update-value "effaçable"

```

Voici le résultat de la commande :

```

root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml set \
  --update-path
"/interfaces/interface[name=Ethernet1]/config/description" \
  --update-value "effaçable"
{

```

```
"source": " 192.168.121.101:6030",
"timestamp": 1781251186020690207,
"time": "2026-06-12T09:59:46.020690207+02:00",
"results": [
  {
    "operation": "UPDATE",
    "path":
"/interfaces/interface[name=Ethernet1]/config/description"
  }
]
```

J'observe bien la paire clé/valeur : "operation":"UPDATE" qui indique que la description a été modifiée avec succès.

Voici la commande pour supprimer la description :

```
gnmic --config gnmic.yaml set \
  --delete
"/interfaces/interface[name=Ethernet1]/config/description"
```

Le même résultat pour supprimer la description :

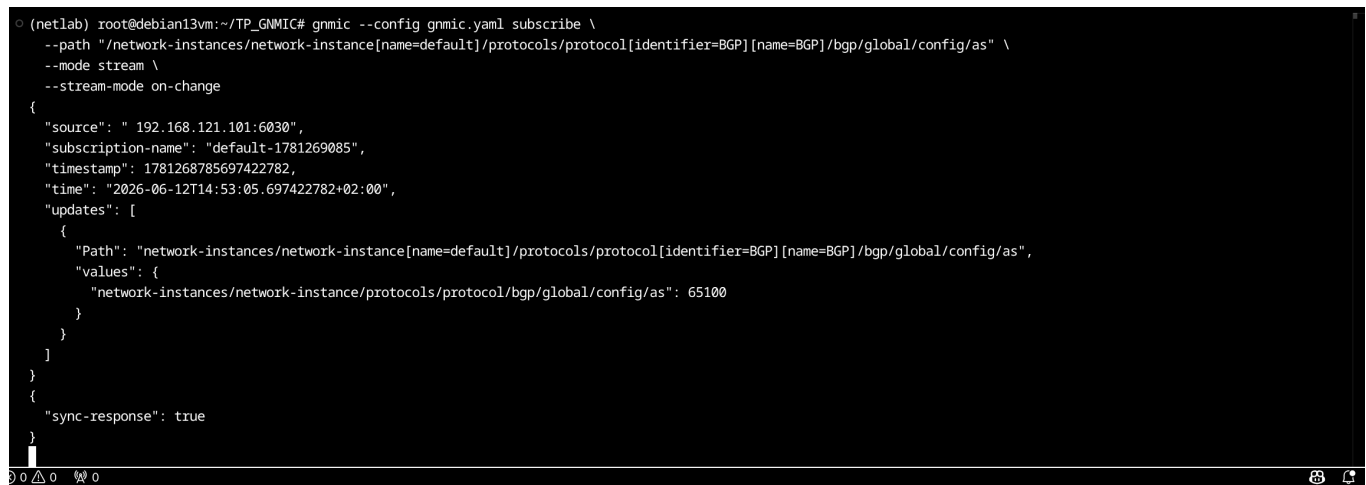
```
root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml set \
  --delete
"/interfaces/interface[name=Ethernet1]/config/description"
{
  "source": " 192.168.121.101:6030",
  "timestamp": 1781251253792139752,
  "time": "2026-06-12T10:00:53.792139752+02:00",
  "results": [
    {
      "operation": "DELETE",
      "path":
"/interfaces/interface[name=Ethernet1]/config/description"
    }
  ]
}
```

```
}
```

15. Utiliser l'événement `on_change` pour être prévenu de la suppression de l'AS BGP.

```
gnmic --config gnmic.yaml subscribe \  
  --path "/network-instances/network-  
instance[name=default]/protocols/protocol[identifiant=BGP]  
[name=BGP]/bgp/global/config/as" \  
  --mode stream \  
  --stream-mode on-change
```

Voici le résultat :



```
○ (netlab) root@debian13vm:~/TP_GNMIC# gnmic --config gnmic.yaml subscribe \  
  --path "/network-instances/network-instance[name=default]/protocols/protocol[identifiant=BGP]  
[name=BGP]/bgp/global/config/as" \  
  --mode stream \  
  --stream-mode on-change  
{  
  "source": " 192.168.121.101:6030",  
  "subscription-name": "default-1781269085",  
  "timestamp": 1781268785697422782,  
  "time": "2026-06-12T14:53:05.697422782+02:00",  
  "updates": [  
    {  
      "Path": "network-instances/network-instance[name=default]/protocols/protocol[identifiant=BGP]  
[name=BGP]/bgp/global/config/as",  
      "values": {  
        "network-instances/network-instance/protocols/protocol/bgp/global/config/as": 65100  
      }  
    }  
  ]  
}  
{  
  "sync-response": true  
}
```

Lorsque l'AS BGP sera supprimée, un message sera affiché dans le terminal.

16. Faites un “shut” de l'interface Ethernet1 et un “no shut”.

Voici la commande pour shutdown une interface (par exemple, eth1) sur le routeur :

```
gnmic --config gnmic.yaml set \  
  --update-path  
"/interfaces/interface[name=Ethernet1]/config/enabled" \  
  --update-value "false"
```

Voici la commande pour no shutdown l'interface eth1 sur le routeur :

```
gnmic --config gnmic.yaml set \
  --update-path
"/interfaces/interface[name=Ethernet1]/config/enabled" \
  --update-value "true"
```

Et la commande pour vérifier à chaque manipulation :

```
gnmic --config gnmic.yaml get \
  --path "/interfaces/interface[name=Ethernet1]/state/admin-
status"
```

5. Gnmi et Prometheus

Voici la procédure d'installation de Prometheus :

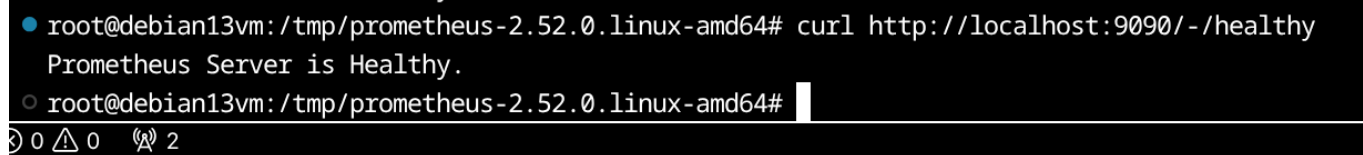
```
cd /tmp
wget
https://github.com/prometheus/prometheus/releases/download/v2.52
.0/prometheus-2.52.0.linux-amd64.tar.gz
tar xzf prometheus-2.52.0.linux-amd64.tar.gz
cd prometheus-2.52.0.linux-amd64

# Lancement de Prometheus
./prometheus --config.file=/root/TP_GNMIC/prometheus.yml &
```

J'ai vérifié l'installation à l'aide de la commande suivante :

```
curl http://localhost:9090/-/healthy
```

```
● root@debian13vm:/tmp/prometheus-2.52.0.linux-amd64# curl http://localhost:9090/-/healthy
Prometheus Server is Healthy.
○ root@debian13vm:/tmp/prometheus-2.52.0.linux-amd64#
```



J'ai également modifié le fichier `gnmic.yml` :

```
cat > gnmic.yml << 'EOF'
address: "192.168.121.101:6030"
username: admin
password: admin
insecure: true
log-file: /tmp/gnmic.log

subscriptions:
  if_counters:
    paths:
      - "/interfaces/interface/state/counters"
    stream-mode: sample
    sample-interval: 10s
    encoding: json_ietf

  bgp_state:
    paths:
      - "/network-instances/network-
instance[name=default]/protocols/protocol[identifier=BGP]
[name=BGP]/bgp/neighbors/neighbor/state"
    stream-mode: sample
    sample-interval: 30s
    encoding: json_ietf

  system_memory:
    paths:
      - "/system/memory/state"
    stream-mode: sample
    sample-interval: 30s
    encoding: json_ietf

outputs:
  prom_output:
    type: prometheus
    listen: ":9273"
    path: /metrics
```

```
    expiration: 60s
EOF
```

Et j'ai créé un fichier `prometheus.yml` :

```
cat > prometheus.yml << 'EOF'
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: "gnmic_ceos1"
    static_configs:
      - targets: ["localhost:9273"]
    metrics_path: /metrics
EOF
```

Après avoir démarré tous les services, voici ce qu'il se passe lorsque je me rends sur l'endpoint de Prometheus :

< > ↻ Not secure 10.202.8.2:9273/metrics

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```
# HELP interfaces_interface_state_counters_carrier_transitions gNMiC generated metric
# TYPE interfaces_interface_state_counters_carrier_transitions untyped
interfaces_interface_state_counters_carrier_transitions{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 4
interfaces_interface_state_counters_carrier_transitions{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 4
interfaces_interface_state_counters_carrier_transitions{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 3
# HELP interfaces_interface_state_counters_in_broadcast_pkts gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_broadcast_pkts untyped
interfaces_interface_state_counters_in_broadcast_pkts{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 4
interfaces_interface_state_counters_in_broadcast_pkts{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_broadcast_pkts{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 0
# HELP interfaces_interface_state_counters_in_discards gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_discards untyped
interfaces_interface_state_counters_in_discards{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_discards{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_discards{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 2
# HELP interfaces_interface_state_counters_in_errors gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_errors untyped
interfaces_interface_state_counters_in_errors{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 1
interfaces_interface_state_counters_in_errors{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 1
interfaces_interface_state_counters_in_errors{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 0
# HELP interfaces_interface_state_counters_in_fcs_errors gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_fcs_errors untyped
interfaces_interface_state_counters_in_fcs_errors{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_fcs_errors{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_fcs_errors{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 0
# HELP interfaces_interface_state_counters_in_multicast_pkts gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_multicast_pkts untyped
interfaces_interface_state_counters_in_multicast_pkts{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 53
interfaces_interface_state_counters_in_multicast_pkts{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_multicast_pkts{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 0
# HELP interfaces_interface_state_counters_in_octets gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_octets untyped
interfaces_interface_state_counters_in_octets{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 19079
interfaces_interface_state_counters_in_octets{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_octets{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 79722
# HELP interfaces_interface_state_counters_in_pkts gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_pkts untyped
interfaces_interface_state_counters_in_pkts{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 124
interfaces_interface_state_counters_in_pkts{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_pkts{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 891
# HELP interfaces_interface_state_counters_in_unicast_pkts gNMiC generated metric
# TYPE interfaces_interface_state_counters_in_unicast_pkts untyped
interfaces_interface_state_counters_in_unicast_pkts{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 67
interfaces_interface_state_counters_in_unicast_pkts{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_in_unicast_pkts{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 891
# HELP interfaces_interface_state_counters_out_broadcast_pkts gNMiC generated metric
# TYPE interfaces_interface_state_counters_out_broadcast_pkts untyped
interfaces_interface_state_counters_out_broadcast_pkts{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_out_broadcast_pkts{interface_name="Ethernet2",source="192.168.121.101:6030",subscription_name="if_counters"} 0
interfaces_interface_state_counters_out_broadcast_pkts{interface_name="Management0",source="192.168.121.101:6030",subscription_name="if_counters"} 0
# HELP interfaces_interface_state_counters_out_discards gNMiC generated metric
# TYPE interfaces_interface_state_counters_out_discards untyped
interfaces_interface_state_counters_out_discards{interface_name="Ethernet1",source="192.168.121.101:6030",subscription_name="if_counters"} 0
```

Je constate que toutes les données ont été adaptées à Prometheus et sont désormais utilisables dans le cadre d'une stack d'observabilité classique Grafana et Prometheus.